

TECHCORP

Custom IAM Solution Design

User Lifecycle Management & Access Control Mechanisms

Prepared as an IAM Developer solution proposal

1. Executive Summary

TechCorp Enterprises operates across more than 100 countries with over 150,000 employees and is expanding its digital footprint through new software systems, cloud services and data repositories. At this scale, manual identity administration and inconsistent access controls create unnecessary security risk, operational overhead and user friction.

This proposal introduces a centralized, automated Identity and Access Management (IAM) model built around two priorities: (1) automated user lifecycle management and (2) stronger, role-based access control. The design combines an authoritative HR source, centralized identity provider, automated provisioning/de-provisioning, RBAC, least privilege, MFA, SSO, privileged access controls, audit logging and periodic access reviews.

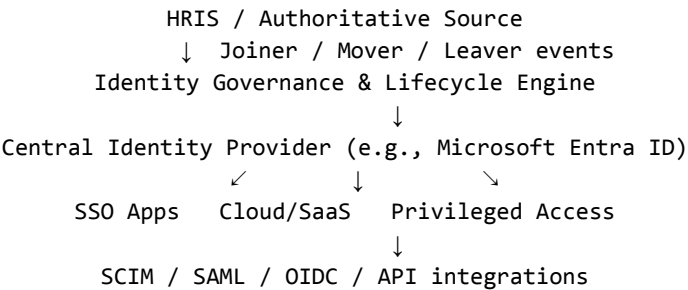
The proposed architecture is scalable and vendor-flexible. Microsoft Entra ID is used as a practical reference implementation for the identity provider, while standards such as SAML 2.0, OpenID Connect (OIDC), SCIM and OAuth 2.0 enable integration with existing and future applications.

2. TechCorp Requirements and Design Goals

Requirement	Proposed Capability	Expected Outcome
Fast onboarding	HR-driven automated account and entitlement provisioning	Employees receive approved access quickly
Secure offboarding	Automated disablement and de-provisioning	Reduced risk from orphaned accounts
Scalable access management	Centralized RBAC and group-based assignment	Consistent control across 150,000+ users
Least privilege	Role/attribute-based entitlements with approvals	Reduced excessive access
Strong authentication	MFA + conditional/risk-based access	Lower credential-compromise risk
Cloud integration	SSO + SCIM + modern federation protocols	Simpler adoption of digital services
Governance	Audit logs, access reviews and reporting	Improved accountability and compliance

3. Target IAM Architecture

The target architecture establishes the HR system as the authoritative source for workforce identity attributes. Identity events flow into a centralized identity platform, which applies lifecycle rules and provisions access to connected applications.



4. Solution A: Enhanced User Lifecycle Management

4.1 Design Overview

TechCorp should implement an automated Joiner–Mover–Leaver (JML) process. The HR system becomes the system of record for employee identity data such as employee ID, department, job title, manager, country, employment status and start/end date. IAM policies consume these attributes to determine account state and appropriate access.

4.2 Joiner / Onboarding Workflow

1. HR creates the employee record and assigns the required organizational attributes.
2. The IAM platform detects the new identity and validates mandatory attributes.
3. A unique digital identity is created in the central directory according to naming and identity-proofing rules.
4. Baseline access is automatically assigned based on department, employment type, country and role.
5. Application-specific access requiring elevated privilege is routed through an approval workflow.
6. The user receives SSO access and completes MFA enrollment before accessing protected services.
7. The transaction is logged for auditability.

4.3 Mover / Role-Change Workflow

8. HR updates the employee's department, title, manager, location or employment attributes.
9. IAM detects the change and recalculates the user's role and entitlements.
10. Access no longer justified by the new role is automatically removed or flagged for approval.
11. New access is granted according to the new role, with approval required for sensitive resources.
12. Conflicting permissions are checked before activation.
13. The old and new entitlement states are recorded for audit and investigation.

4.4 Leaver / Offboarding Workflow

14. HR records the employee's termination date/status.
15. At the effective time, the central identity is disabled immediately or according to the approved termination policy.
16. Active sessions and refresh tokens are revoked where supported.
17. Application accounts and entitlements are de-provisioned through SCIM/API integrations.
18. Privileged credentials, VPN access, badges and other linked access are revoked through integrated workflows.
19. Ownership of business data is transferred to the approved manager or data owner.
20. A complete offboarding record is retained for audit.

4.5 Technologies and Standards

- Microsoft Entra ID (reference IdP/directory) or an equivalent enterprise identity platform.
- HRIS integration using REST APIs, event-driven workflows or supported connectors.
- SCIM 2.0 for automated user and group provisioning to compatible SaaS applications.
- SAML 2.0 and OpenID Connect for federated SSO.
- OAuth 2.0 for delegated authorization where appropriate.
- Identity Governance / access-review capabilities for entitlement certification.

- SIEM integration for centralized security monitoring and audit analysis.

4.6 Lifecycle Controls

- No employee should retain access solely because an administrator forgot to remove it.
- Access should be driven by authoritative employment and organizational attributes.
- High-risk entitlements should require explicit approval rather than automatic assignment.
- Dormant, orphaned and inactive accounts should be detected and investigated.
- Service and non-human identities should follow a separate lifecycle policy with named owners and credential rotation.

5. Solution B: Strengthened Access Control Mechanisms

5.1 RBAC and Least Privilege Model

TechCorp should use RBAC as the primary authorization model, with attributes used to refine access where roles alone are insufficient. Roles should represent business responsibilities rather than individual users. Permissions are attached to roles, and users receive roles through controlled group membership or entitlement workflows.

Example Role	Typical Access	Approval	Risk Level
HR Analyst	HR applications and approved employee records	Manager + data owner for sensitive data	Medium
Software Developer	Source repositories, development environments	Manager; elevated production access separately	Medium
Finance Analyst	Finance applications and reporting data	Manager + finance owner	High
System Administrator	Infrastructure administration	Privileged access workflow	Critical

5.2 MFA and Conditional Access

MFA should be mandatory for workforce identities and enforced more strongly for sensitive applications. A conditional-access layer should evaluate signals such as device compliance, location, sign-in risk, application sensitivity and authentication strength.

- Require phishing-resistant MFA (for example, FIDO2/security keys or passkeys) for privileged and high-risk access where feasible.
- Require MFA for access to sensitive corporate resources and remote access.
- Block or challenge suspicious sign-ins based on risk signals.
- Require managed/compliant devices for high-value internal applications where appropriate.
- Apply stricter policies to administrator accounts than standard workforce accounts.

5.3 Privileged Access Management (PAM)

Administrative access should not be permanently assigned where it can be avoided. Privileged roles should use just-in-time elevation, approval, strong authentication, session logging and time-bound permissions. This limits the blast radius of compromised administrator credentials.

5.4 Access Reviews and Segregation of Duties

- Managers and resource owners periodically certify that users still require assigned access.
- Sensitive and privileged entitlements should be reviewed more frequently than ordinary access.
- Segregation-of-duties rules should prevent incompatible combinations, such as initiating and approving the same financial transaction.
- Unapproved or stale access should be automatically revoked or routed to remediation.

5.5 Audit and Monitoring

All significant identity and access events should be centrally logged, including authentication attempts, MFA events, privilege elevation, role changes, provisioning, de-provisioning and access-policy decisions. Logs should feed the enterprise SIEM to support detection, investigation and compliance reporting.

6. Alignment with TechCorp Business Processes

Business Process	IAM Integration	Operational Benefit
Recruitment & onboarding	HR event automatically starts identity creation and baseline access	Faster employee readiness
Internal transfers	HR attribute changes trigger role recalculation	Access follows responsibility
Termination	Termination event triggers automated de-provisioning	Rapid risk reduction
Cloud/SaaS adoption	SSO and SCIM connect applications to central IAM	Reduced account sprawl
IT support	Self-service password/MFA processes and centralized identity	Fewer repetitive tickets
Compliance	Access reviews and immutable audit trails	Easier evidence collection

7. Alignment with TechCorp Business Objectives

Security: Centralized authentication, least privilege, MFA, PAM and automated offboarding reduce common identity-related attack paths.

Efficiency: Automation removes repetitive provisioning tasks, reduces manual errors and accelerates employee access.

User Experience: SSO reduces password fatigue and provides a consistent access experience across corporate and cloud applications.

Scalability: Role/group-based access and standards-based integrations allow the model to support growth across 100+ countries and 150,000+ employees.

Competitive Edge: A secure and frictionless digital workplace allows TechCorp to adopt cloud services and new applications without creating uncontrolled identity sprawl.

8. Rationale for Key Design Decisions

Decision	Rationale
Centralized identity provider	Creates a single control plane for authentication, policy enforcement and identity visibility while supporting SSO.
HR as authoritative source	Employment status and organizational attributes are maintained by the business system responsible for them, reducing conflicting identity records.
JML automation	Removes dependence on manual tickets and administrator memory, especially important at TechCorp's scale.
RBAC	Maps technical permissions to business responsibilities and makes access easier to administer consistently.
Least privilege + approvals	Ensures users receive only the access justified by their role and that sensitive permissions receive additional scrutiny.
MFA + conditional access	Adds protection when passwords are compromised and allows risk-sensitive access decisions.
SCIM / SAML / OIDC	Uses established standards to integrate SaaS and enterprise applications while reducing custom integration effort.
PAM	Limits standing administrative privilege and reduces the impact of compromised privileged credentials.
Audit + SIEM	Provides traceability and supports detection, investigation and compliance evidence.

9. Implementation Roadmap

Phase	Key Activities
Phase 1: Foundation	Inventory identities, applications and privileged accounts; establish governance, ownership and identity data standards.
Phase 2: Identity Integration	Connect HRIS to the identity platform; establish JML workflows and identity matching rules.
Phase 3: Core Access	Implement RBAC, groups, SSO and MFA; onboard high-priority applications.
Phase 4: Governance	Introduce access reviews, SoD controls, PAM and automated remediation.
Phase 5: Scale & Optimise	Expand application integrations globally, tune policies, automate reporting and continuously improve based on metrics.

10. Success Metrics

- Percentage of joiners provisioned automatically before their start date.
- Average time from HR termination event to identity disablement.

- Percentage of connected applications using SSO and automated provisioning.
- Percentage of users protected by MFA and stronger authentication methods.
- Number and age of orphaned/dormant accounts.
- Percentage of access reviews completed on time.
- Number of excessive or conflicting entitlements identified and remediated.
- Reduction in IAM-related service desk tickets and manual provisioning tasks.

11. Risks and Mitigations

Risk	Impact	Mitigation
Poor HR data quality	Incorrect access assignment	Define mandatory identity attributes, validation and ownership.
Legacy applications lack standards support	Manual provisioning remains	Use APIs/connectors or controlled gateway workflows; prioritise modernization.
Overly broad roles	Excess privilege	Perform role mining, approval workflows and periodic access reviews.
User resistance to MFA	Adoption friction	Use SSO, clear communication and user-friendly phishing-resistant methods.
Global regulatory differences	Compliance exposure	Apply a global baseline with country-specific policy overlays where legally required.

12. Conclusion

The proposed IAM design gives TechCorp a scalable identity foundation capable of supporting a global workforce and rapidly expanding digital environment. Automated JML processes ensure that identities and access change with employment status and responsibilities, while RBAC, least privilege, MFA, PAM and continuous monitoring protect critical resources. By integrating IAM into existing HR, IT and business workflows rather than treating it as a standalone security layer, TechCorp can improve security and operational efficiency simultaneously.

The recommended approach is deliberately standards-based and scalable. It can be implemented incrementally, starting with the highest-risk applications and identity processes, then expanded across TechCorp's global environment as governance and integrations mature.